THE BUSINESS ANALYST
THE BRIDGE BETWEEN WHY AND HOW

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BIO

Martina Babini is the Principal Business Analyst of LuxExperience's Technology.

In recent years, she has worked as a Business Architect and Principal Analyst of YOOX NET-A-PORTER GROUP, and then moved to this role a year ago after the acquisition of YOOX NET-A-PORTER GROUP.

LuxExperience B.V. is a leading global digital multi-brand luxury group formed by the merger of prominent fashion platforms, including Mytheresa, NET-A-PORTER, MR PORTER, and YOOX.

Lecture Overview: Strategic Roles & Methodologies

This lecture explores the strategic role of the **Business Analyst and Architect** within the e-commerce sector, focusing on the integration of Scrum and data-driven decision-making.

Key Areas of Focus:

- **Dual Case Study:** Optimizing Solr infrastructures and managing M&A integrations.
- **Requirements Lifecycle:** Examining end-to-end processes in complex environments.
- **Stakeholder Management:** Utilizing the *'5 Whys'* technique to uncover root business needs.
- **System Evolution:** Applying Martin Fowler's *'Transitional Architecture'*.

Business Analyst

The Core Mission

To identify business problems and design technical solutions that deliver **actual value**.

They ensure the engineering team builds the **right** product, not just a **functioning** product.

MAIN RESPONSIBILITIES

Requirements Engineering

Eliciting, documenting, and managing **functional** (what the system should do) and **non-functional** requirements.

- Expected performances
- Response times
- And more...

Gap Analysis

Analyzing the **"As-Is"** state (current system) vs. the **"To-Be"** state (proposed system).

Goal: To identify the set of changes and highlight:

- Potential risks
- Critical issues
- Strategic alignment

Stakeholder Management

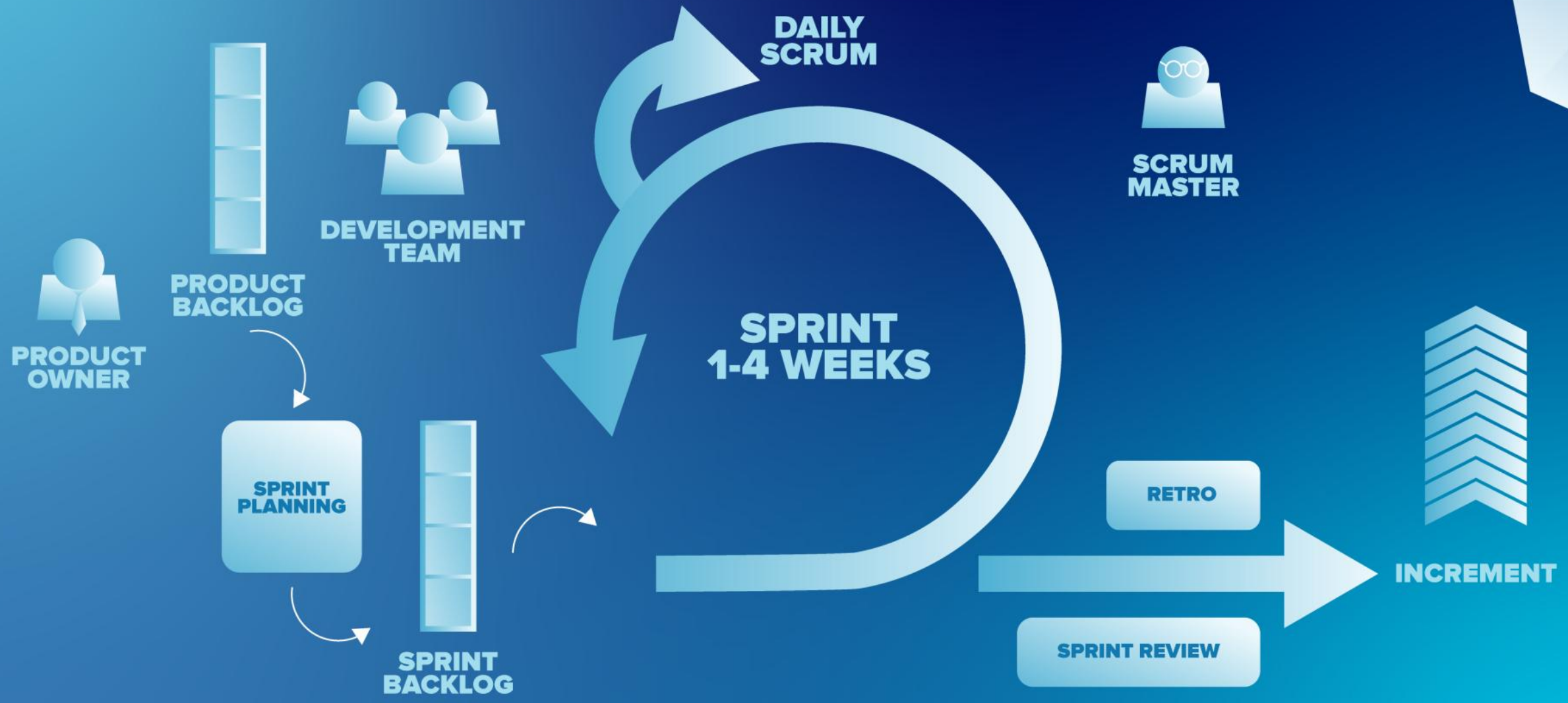
Navigating the **"user-error"** of vague human requests to create precise technical specifications.

Value Validation

Testing if the final build actually solves the **original business pain point**.

SCRUM & BA ROLE

SCRUM FRAMEWORK



THE ROLE OF A BUSINESS ANALYST (BA) IN SCRUM

In the official *Scrum Guide*, the "Business Analyst" is not listed as a formal role—the framework only defines the **Product Owner**, the **Scrum Master**, and the **Developers**.

However, in modern practice, BAs are essential bridge-builders.

Part of the Developers

Collaborates directly on building increments and defining technical requirements.

PO's Strategic Right Hand

Supports the Product Owner in backlog refinement and stakeholder communication.

BA AND PO

Feature	Business Analyst	Product Owner
Primary Focus	Requirements, logic, and "How it works."	Value, ROI, and "What we build next."
Authority	Recommends and documents.	Decides and prioritizes.
Daily Activity	Deep dives into user stories and edge cases.	Communicating vision and managing the roadmap.

THE BA TOOLKIT: STRATEGIC TECHNIQUES BY OBJECTIVE

DISCOVERY & ROOT CAUSE ANALYSIS

Goal

To understand the "Why" and identify the core problem before jumping to solutions.

5 Whys

Iterative interrogation to peel back layers of symptoms.

Fishbone (Ishikawa) Diagram

Categorizing potential causes of a problem.

Root Cause Analysis (RCA)

Systematic process for identifying the origin of an issue.

REQUIREMENTS ELICITATION

Goal

To gather comprehensive information from stakeholders and users.

Stakeholder Interviews

One-on-one deep dives to uncover specific needs.

Workshops

Collaborative sessions to align multiple perspectives quickly.

Observation (Job Shadowing)

Watching users in their natural environment to find "hidden" pain points.

PROCESS & LOGIC MODELING

Goal

To visualize complex workflows and ensure technical feasibility.

BPMN (Business Process Model and Notation)

Mapping out "As-Is" vs. "To-Be" workflows.

User Story Mapping

Organizing tasks into a visual journey to plan releases.

Data Flow Diagrams (DFD)

Visualizing how information moves through a system.

STRATEGY & PRIORITIZATION

Goal

To align the project with business goals and manage the backlog.

SWOT Analysis

Assessing Strengths, Weaknesses, Opportunities, and Threats.

MoSCoW Prioritization

Categorizing requirements into Must-have, Should-have, Could-have, and Won't-have.

Gap Analysis

Determining steps needed to move from the current state to the future state.

5 WHYS EXAMPLE

ROOT CAUSE ANALYSIS: 5 WHYS METHOD

Objective: Reduce Cart Abandonment via AI Chatbot Integration

1. Why think a chatbot reduces abandonment?	**Stakeholder:** Customers leave the checkout to find information and never return.
2. Why leave to find information?	**BA Investigation:** Statistics show users click away specifically at the "Discount Code" box.
3. Why click away at the Discount box?	**User Interviews:** Users search Google for promo codes to avoid overpaying.
4. Why feel they are overpaying?	**Competitive Analysis:** Prices are 15% higher due to "coupon culture" strategy.
5. Why is the box so prominent?	**Root Cause:** UI template was built for "Flash Sales" where codes are mandatory.

DATA-DRIVEN VALIDATION: MOVING FROM INTUITION TO EVIDENCE

BASELINE METRICS (THE "AS-IS" STATE)

Goal

Quantify the current pain point to prove the need for change.

Process Cycle Time

Measuring how long a task currently takes to identify bottlenecks.

Error Rates

Tracking the frequency of manual failures or rework.

Volume Analysis

Identifying peak usage times or high-traffic features to focus development efforts where they matter most.

PREDICTIVE ANALYSIS & ESTIMATION

Goal

Support the proposal by forecasting the potential return on investment (ROI).

Trend Analysis

Using historical data to project future growth and system capacity needs.

Cost-Benefit Analysis

Comparing the data-backed cost of implementation against the projected savings in time or resources.

What-If Scenarios

Running data models to see how different variables (e.g., a 10% increase in users) affect the proposed solution.

POST-IMPLEMENTATION VALIDATION

Goal

Close the feedback loop by proving the solution actually worked.

A/B Testing

Comparing the proposed solution against the current version to see which yields better user behavior data.

KPI Tracking

Monitoring Key Performance Indicators (e.g., Conversion Rate, CSAT) against the original baseline.

User Analytics

Analyzing click-streams and heatmaps to see if users are interacting with the new feature as intended.

CASE STUDY – OPTIMIZING PERFORMANCE THROUGH DATA INSIGHTS

SOLVING SOLR PERFORMANCE ISSUES

Strategic Language Localization Case Study

The Problem: Business Complaint: The website’s product catalog generation was unacceptably slow.

Technical Context

Solr: Open-source enterprise search platform. Acts as the "search engine" powering product searches, filters, and categories.

The Index: Creates a structured map of products. More indices directly increase "build time" (generation).

The Setup

The catalog was indexed by **Country** and **Language**.

English and Chinese were active in all **172 countries**.

THE BA DATA-DRIVEN HYPOTHESIS

Questioning the "Standard" Configuration

The Challenge: Does every country actually need a Chinese language index?

Analysis Framework

Net Sales: Total revenue minus returns/discounts. The actual money kept.

The Goal: Identify "dead weight" countries (high cost, zero/low revenue).

The Analysis Method

- **Metric A:** Order volume from Chinese-navigating users.
- **Metric B:** Total Net Sales from those specific orders.
- **Comparison:** Technical cost (build time) vs. financial gain.

RESULTS & BUSINESS IMPACT

Data-Backed Optimization Results

THE OUTCOME

Rationalization: The analysis proved that Chinese was only driving significant revenue in **10 out of 172 countries**.

The Proposal: Remove the Chinese index for the remaining **162 countries**.

Efficiency Gain

Catalog generation time was reduced by **1 full hour**.

Resource Optimization

Significant reduction in **server load** and processing power requirements.

User Experience

Faster updates to web catalog, ensuring products **reach market quicker**.

THE BA AT ENTERPRISE LEVEL

UNPACKING THE ROLES: **A COMPARATIVE VIEW**

	SCRUM/TEAM BA	ENTERPRISE BA	BUSINESS ARCHITECT
Horizon	Tactical (Weeks/Months)	Project Portfolios & ROI	Business Models & Strategy
Focus	User Stories & Sprints	Operational (Quarterly)	Strategic (1-3+ Years)
Stakeholders	Developers & Product Owner	Department Heads	C-Suite (CEO, CTO, COO)
Key Question	"How do we build this feature?"	"Which projects should we fund?"	"What capabilities do we need to win?"

THE M&A CHALLENGE – UNIFYING TWO WORLDS

STRATEGIC INTEGRATION OF TWO EQUAL-SIZED ENTITIES

The Scenario

Context: Company A and Company B are merging. Both have established, complex ecosystems.

The Mission

Transition from two fragmented landscapes into a single, unified Enterprise Architecture.

The Architect’s Primary Objective

Ensure **"Business Continuity"** while driving synergy and reducing redundant operational costs.

STEP 1: THE LANDSCAPE AUDIT & GAP ANALYSIS

Inventory of Assets

- **ERP:** Finance, HR, and Supply Chain.
- **OMS:** Capturing and fulfilling orders.
- **Front-ends:** Web, Mobile, and POS systems.

Gap Analysis

Overlaps: Identifying redundancies like duplicate SAP instances.

Gaps: Identifying missing capabilities (e.g., loyalty engines).

Outcome: The Heat Map

A visual guide showing which capabilities are redundant, superior, or missing.

STEP 2: DEFINING THE TARGET STATE

Designing the "To-Be" Architecture

Keep (Best of Breed)

Selecting the superior system (e.g., Company B's OMS is more scalable).

Retire

Decommissioning systems that are redundant or technically obsolete.

Merge/Replace

Implementing a new, shared cloud-native solution for both entities.

The Goal

A unified architectural blueprint that supports the combined business volume without doubling maintenance costs.

STEP 3: THE TRANSITIONAL ARCHITECTURE

The Risk of the "Big Bang"

In E-commerce, switching everything overnight is extremely high-risk.

Consequences:
Data loss, checkout downtime, and broken supply chains.

The Martin Fowler Approach

Incremental Migration:
We do not move everything at once. We define **Transitional States**.

Transitional States Defined

Intermediate architectures where parts of Company A and Company B coexist and communicate seamlessly.

THE STRANGLER FIG PATTERN

SEAMLESS REPLACEMENT OF LEGACY SYSTEMS

Execution

We build new functionality (the "To-Be" system) around the edges of the old system (the "Legacy").

Analogy:
Named after the vine that grows around a tree, eventually replacing it.

Interception

A "Facade" or "Proxy" redirects traffic. New features go to the new system, while old features remain in legacy until ready to migrate.

Final Result

The legacy system is "strangled" and can be safely decommissioned once it no longer handles any traffic.

DESIGNING FOR CHANGE: BRIDGES & ACLS

PROTECTING THE INTEGRITY OF THE NEW SYSTEM

The Problem

When Company A's new system talks to Company B's old, messy database, the "mess" often leaks into the new design.

The Solution

The Architect designs an **Anti-Corruption Layer (ACL)**—a temporary "bridge" or translation layer.

Function & Purpose

It translates semantics between systems so the architecture remains "clean" and avoids inheriting legacy technical debt during transition.

MIGRATION STRATEGY BENEFITS

WHY WE USE TRANSITIONAL ARCHITECTURES

Business Continuity

Seamless user experience during transition. Customers should never know a migration is happening.

De-risking

Using the **Strangler Fig Pattern** to provide value early and often.

Avoids high-risk "Go-Live" bottlenecks.

Strategic Agility

The **ACL** ensures the Target Architecture is built on modern standards, not compromised by legacy debt.

Key Objective

Balancing technical excellence with business continuity ensures a successful transition without operational disruption.

WHAT ABOUT AI?